

Intrathecal chemotherapy

Chemotherapy given into the fluid around the brain and spinal cord is called intrathecal chemotherapy. It is given during a lumbar puncture.

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What is intrathecal chemotherapy?

Sometimes doctors give [chemotherapy](#) into the fluid around the spinal cord during a lumbar puncture. This is called intrathecal chemotherapy.

What is a lumbar puncture?

A [lumbar puncture](#) can be done in an outpatients department or on a ward at the hospital.

A doctor or specialist nurse will do the lumbar puncture. Your doctor or nurse will usually give you a local anaesthetic in your lower back first. Then they will insert a needle between 2 of the spinal bones. The doctor or nurse then takes a sample of cerebrospinal fluid (CSF) for testing.

What is cerebrospinal fluid (CSF)?

Cerebrospinal fluid is known as CSF. It is a watery fluid which surrounds the brain and spinal cord. The CSF protects the brain and spine from injury.

When is intrathecal chemotherapy used?

In some cancers, such as some leukaemias or lymphomas, the cancer cells can pass into the CSF. Chemotherapy given into a vein or by mouth cannot easily get into the CSF. Intrathecal chemotherapy may be given if a cancer has spread to the CSF, or if there is a risk it could.

Only certain chemotherapy drugs can be given in this way. The chemotherapy drugs that are most commonly given intrathecally (with a lumbar puncture) are:

- [methotrexate](#)
- [cytarabine](#).

[Steroids](#) can also be given this way.

Before having intrathecal chemotherapy

Before your treatment, your doctor will explain why you need it. They should give you information about:

- the lumbar puncture
- what will be done
- the chemotherapy treatment
- any other tests or treatments that may be available
- any risks or side effects of the lumbar puncture and chemotherapy.

You will be asked to sign a form saying that you [give your permission \(consent\)](#) for the hospital staff to do the lumbar puncture and give the intrathecal chemotherapy.

Specially trained staff will give the treatment in a room in the hospital which is only used for intrathecal chemotherapy. Very rarely, you may have to travel to a different hospital for this treatment.

It might help to have a family member or friend with you when the procedure is explained, to help you remember what was said. You may also find it useful to write down a list of questions before you go to your appointment.

If you do not understand what you have been told, tell the staff straight away so they can explain again. Some medical treatments and procedures can be difficult to understand. It is not unusual to need things explained again.

People sometimes feel that doctors and nurses may be too busy to answer their questions. But it is important for you to know how the intrathecal chemotherapy is likely to affect you.

Having intrathecal chemotherapy

A nurse or doctor will check your blood pressure and pulse before the treatment. You will need to lie flat for some time after the procedure, so you might want to go to the toilet first.

During your lumbar puncture, you will either:

- lie on one side with your knees pulled up towards your chest

Image: This illustration shows a person having a lumbar puncture, which involves lying on one side with the knees drawn up to the chest. Local anaesthetic is injected to numb the area. A hollow needle is then inserted between two of the spinal bones and into the spinal canal. If you're having intrathecal chemotherapy, the drugs will be given into the fluid around the spinal cord during the lumbar puncture.

- sit up, bending forwards over a table and supported by pillows.

Image: This illustration shows a person having a lumbar puncture. They are sitting up and leaning over a table, with their back towards us. Their elbows are resting on the table. It also shows the hands of a doctor holding a syringe. The needle of the syringe is positioned over the middle of the lower part of the person's back (where the spine is).

These positions allow your back to curve as much as possible so that there are small spaces between the bones of your spine (vertebrae).

The doctor or nurse will clean the skin over the lower part of the back with an antiseptic solution. They will then inject a local anaesthetic to numb the area. This may take a few minutes to work.

They will then insert a needle between 2 of the spinal bones and into the spinal canal.

You will need to stay very still while they do this. You may feel some pressure as the needle is put in. The doctor or nurse will:

- take a sample of CSF
- inject chemotherapy drugs
- do both.

There are very strict guidelines for giving chemotherapy in this way. The doctor and nurses will do a number of safety checks. This is to make sure you are only given the medicines that have been prescribed for you. There may be extra nurses or doctors in the room to do the safety checks. If you would like to, you can also check the drugs before they are given.

After the drug has been given, the doctor or nurse will remove the needle and place a small dressing over the injection area. The whole process normally takes around 20 minutes.

If you are having intravenous chemotherapy (into a vein) on the same day, you will have it first in the ward or day unit. The intrathecal chemotherapy will only be given after the other chemotherapy has finished.

After intrathecal chemotherapy

After the procedure, you will need to lie flat for a while. This may be from 1 hour to a few hours, depending on how you feel.

Having this treatment is not usually painful. Some people may find it uncomfortable or have a headache for a few hours. Tell your doctor or nurse if you have a headache, as they can give you mild painkillers to help. Sitting up can make a headache worse.

You will normally have your blood pressure and pulse checked again during this time. Ask the doctor or nurse when it will be safe for you to sit up.

Once you have rested and feel well, you will be able to get up and move around. But it is best not to drive or operate machinery for 24 hours after a lumbar puncture.

If you are having intrathecal chemotherapy, you may need several doses and will need to have a lumbar puncture each time.

If you have any questions about lumbar punctures or intrathecal chemotherapy, ask your doctor or nurse at the hospital where you are being treated.

About our information

References



Below is a sample of the sources used in our chemotherapy information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Brighton, D. Wood, M. The Royal Marsden Hospital Handbook of Cancer Chemotherapy. Elsevier Churchill Livingstone. 2005.

National Institute for Health and Care Excellence (NICE) Neutropenic Sepsis Guideline CG151. 2012.

Perry, MC. The Chemotherapy Source Book (5th ed.) Philadelphia: Lippincott, Williams & Wilkins. 2012.

UKONS Acute Oncology Initial Management Guidelines Version 3, March 2018. Available from www.ukons.org (accessed June 2021).

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Chief Medical Editor, [Professor Tim Iveson](#), Consultant Medical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 March 2022 | Next review: 01 March 2025

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help